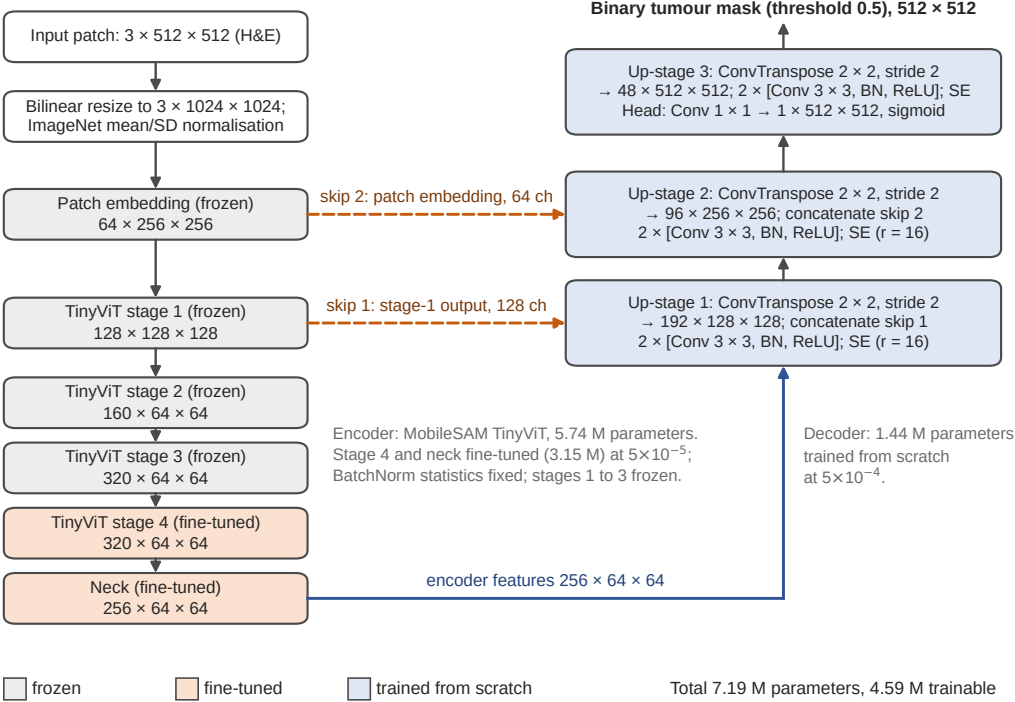
Input patch: 3 × 512 × 512 (H&E)
Bilinear resize to 3 × 1024 × 1024; ImageNet mean/SD normalisation
Patch embedding (frozen) 64 × 256 × 256
TinyViT stage 1 (frozen) 128 × 128 × 128
TinyViT stage 2 (frozen) 160 × 64 × 64
TinyViT stage 3 (frozen) 320 × 64 × 64
TinyViT stage 4 (fine-tuned) 320 × 64 × 64
Neck (fine-tuned) 256 × 64 × 64
skip 2: patch embedding, 64 ch
skip 1: stage-1 output, 128 ch
Encoder: MobileSAM TinyViT, 5.74 M parameters. Stage 4 and neck fine-tuned (3.15 M) at 5×10−5; BatchNorm statistics fixed; stages 1 to 3 frozen.
encoder features 256 × 64 × 64
Up-stage 1: ConvTranspose 2 × 2, stride 2 → 192 × 128 × 128; concatenate skip 1 2 × [Conv 3 × 3, BN, ReLU]; SE (r = 16)
Up-stage 2: ConvTranspose 2 × 2, stride 2 → 96 × 256 × 256; concatenate skip 2 2 × [Conv 3 × 3, BN, ReLU]; SE (r = 16)
Up-stage 3: ConvTranspose 2 × 2, stride 2 → 48 × 512 × 512; 2 × [Conv 3 × 3, BN, ReLU]; SE Head: Conv 1 × 1 → 1 × 512 × 512, sigmoid
Binary tumour mask (threshold 0.5), 512 × 512
Decoder: 1.44 M parameters, trained from scratch at 5×10−4.
frozen
fine-tuned
trained from scratch
Total 7.19 M parameters, 4.59 M trainable